

Secure Virtual Assistant with Speaker Recognition

Final Technical Report

Project: Secure-VA

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1 Tom tat do an

Do an xay dung mot tro ly ao bao mat co kha nang nhan lenh bang giong noi, nhan dien nguoi noi, xac minh nguoi noi truoc cac tac vu nhay cam, va ca nhan hoa phan hoi theo tung nguoi dung. He thong duoc chia thanh hai yeu cau chinh:

- **YC1 - Speaker Recognition:** train va danh gia mo hinh ECAPA-TDNN cho Speaker Identification (SID) va Speaker Verification (SV).
- **YC2 - Secure Virtual Assistant:** tich hop SID/SV vao tro ly ao, phan quyen tac vu theo ba muc NORMAL, PERSONAL va IMPORTANT.

Ban hien tai da co pipeline end-to-end: thu am tren trinh duyét, decode audio, VAD, ASR, speaker embedding, NLU, auth gate, intent handler va TTS. Evidence speaker recognition da duoc tao tren VoxCeleb Indian subset va VIVOS. Checkpoint VIVOS duoc chon lam checkpoint chinh cho demo tieng Viet vi cho ket qua tot hon trong thiet lap noi bo.

2 Pham vi nop bai

Dua tren tai lieu va artifact hien co, goi nop bai nen gom cac muc sau:

Thanh phan	Ly do can nop
core/	Logic dung chung: audio preprocessing, ASR, speaker encoder, database, NLU, router, handlers, audit, OAuth, Gmail, reminder.
web/	Flask web app de demo enrollment, assistant, admin, file manager, music, user management.
cli/	CLI enrollment, assistant va smoke test, huu ich neu demo web gap loi.
training/	Code train/evaluate ECAPA-TDNN, model card, ket qua train/evaluate.
scripts/	Cac script tao split VIVOS, benchmark, calibration, migration va export dataset NLU.
tests/	Unit/integration tests chung minh cac fix bao mat va regression.
requirements.txt va requirements.lock.txt	Moi truong chay ung dung.
.env.example	Mau cau hinh, khong chua secret.
checkpoints/vios/	Checkpoint VIVOS va spk2idx.json dung cho demo. Neu file lon qua, co the nop link Drive.
training/results/	Artifact evidence: split, log train, SID/SV results, checkpoint, summary.
report/main.pdf	Bao cao chinh de nop. File nay duoc tao tu report/main.tex.

Khong nen nop cac muc sau neu public hoac gui cho nguoi khac:

- `.env`: co the chua API key, OAuth secret, Flask secret.
- `venv/`: moi truong ao rat lon, co the cai lai bang requirements.
- `.git/`: metadata git khong can thiet cho cham bai.
- `data/users.db`, `data/enroll_audio/`, `data/user_files/`: chi nop neu giang vien yeu cau demo data mau va da chac chan khong co du lieu rieng tu.
- `docs/`: khong bat buoc neu da nop bao cao PDF nay. Thu muc docs co the giu trong repo noi bo lam evidence phu.

3 Kien truc he thong

3.1 Tong quan pipeline

Mot luot lenh giuong noi duoc xu ly theo chuoii:

1. Browser thu audio bang MediaRecorder va gui len endpoint Flask.
2. Backend decode WebM/Opus thanh audio mono 16 kHz.
3. Silero VAD cat bot khoang lang, giu vung co speech.
4. ASR chuyen audio thanh transcript tieng Viet.
5. Speaker encoder tao embedding 192 chieu.
6. NLU phan loai intent va trich entity.
7. Router doc auth level cua intent va quyet dinh co can SID/SV hay khong.
8. Handler thuc thi tac vu neu pass policy.
9. TTS tao audio phan hoi va gui ve browser.

3.2 Cac module chinh

Module	Vai tro
<code>core/audio_io.py</code>	Decode audio, VAD, preprocess waveform.
<code>core/asr.py</code>	Wrapper cho faster-whisper va PhoWhisper neu cau hinh.
<code>core/speaker_encoder.py</code>	PyGAPPA/WavLM encoder, load checkpoint tu train hoac fallback pretrained.
<code>core/database.py</code>	SQLite user DB, speaker centroid, preferences, reset request, alert.
<code>core/intents.py</code>	Dinh nghia intent va auth level.
<code>core/nlu.py</code>	Gemini function calling va rule-based fallback.
<code>core/router.py</code>	Auth gate: NORMAL, PERSONAL, IMPORTANT.
<code>core/handlers.py</code>	Logic thuc thi tung intent.

Module	Vai tro
core/oauth.py, core/gmail_api.py	Google OAuth va Gmail send.
web/app.py	Flask routes, API, session, CSRF guard, admin dashboard.
training/train_ecapa.py	Train ECAPA-TDNN cho SID.
training/evaluate_sid.py	Evaluate Top-1/Top-5 SID.
training/evaluate_sv.py	Evaluate EER, threshold va minDCF cho SV.

4 Mo hình bảo mật theo intent

Hệ thống không xử lý mọi lệnh như nhau. Mọi intent có một mức bảo mật:

Mức	Ý nghĩa	Ví dụ intent	Policy
NORMAL	Tác vụ công khai, ít rủi ro	get_time, tell_joke, general_question	Không cần xác thực giọng.
PERSONAL	Tác vụ cá nhân hóa	greet, play_music, show_schedule	Dùng SID để biết người nói là ai.
IMPORTANT	Tác vụ nhạy cảm hoặc thay đổi dữ liệu	send_email, read_notes, delete_data, open_files	Cần SV pass trước khi handler chạy.

Thiết kế này giảm rủi ro: người lạ có thể hỏi giờ, nhưng không thể đọc ghi chú, gửi email hay mở file nếu không xác minh đúng giọng.

5 Speaker Recognition

5.1 ECAPA-TDNN

Mô hình speaker chính là ECAPA-TDNN. Cấu hình:

- Feature: 80-dim log Mel filterbank.
- Normalization: cepstral mean normalization theo trục thời gian.
- Embedding dimension: 192.
- Loss: Additive Angular Margin Softmax.
- Margin: 0.2.
- Scale: 30.
- Crop train: 3 giây.
- Verification score: cosine similarity giữa hai embedding đã L2-normalize.

Mô hình được train theo bài toán speaker classification. Sau train, phần encoder được tái sử dụng cho speaker verification bằng cosine similarity.

5.2 Dataset

Hai dataset đã được dùng để tạo evidence:

Dataset	Domain	Speakers	Utterances	Train/Val/Test	Tri
VoxCeleb Indian sub-set	Celebrity speech, Indian/English accent	24	4,857	3,389 / 717 / 751	
VIVOS	Vietnamese read speech	65	12,419	8,686 / 1,850 / 1,883	4,

VIVOS gần domain tiếng Việt của ứng dụng hơn. VoxCeleb Indian được giữ làm baseline để kiểm tra domain mismatch.

5.3 Training setup

Tham số	Giá trị
Epochs	15
Batch size	64
Learning rate	0.001
Optimizer	Adam, weight decay $2e-5$
Scheduler	CosineAnnealingLR
Sample rate	16 kHz
Crop duration	3.0 seconds
Device	CUDA trên Kaggle/GPU
Seed	42

5.4 Kết quả

Dataset	Best Val Acc	SID Top-1	SID Top-5	SV EER	minDCF	Thr@E
VoxCeleb Indian	93.44%	91.34%	98.67%	2.90%	0.1486	0.3
VIVOS	99.19%	99.31%	100.00%	0.96%	0.0673	0.4

VIVOS tốt hơn rõ ràng trên cả SID và SV. SID Top-1 tăng từ 91.34% lên 99.31%. SV EER giảm từ 2.90% xuống 0.96%. Kết quả này ủng hộ việc chọn checkpoint VIVOS làm model chính cho demo tiếng Việt.

5.5 Caveat về metric

Kết quả trên là metric nội bộ tự split tự tạo, không phải benchmark chuẩn VoxCeleb1-O hay NIST SRE. VIVOS là read speech và có thể sạch hơn mọi trường hợp. Trial set được sinh từ tập test nội bộ nên chỉ nên dùng để chứng minh pipeline, so sánh nội bộ và chọn checkpoint.

6 Augmentation va fine-tune noisy

Codebase da co khung augmentation:

- MUSAN noise injection.
- RIR reverb.
- Speed perturbation 0.9x, 1.0x, 1.1x.
- SpecAugment tren log-mel feature.

Trong ban nop nay, fine-tune noisy/reverb chua duoc dua vao metric chinh do gioi han thoi gian va GPU session. Dieu nay khong lam hong do an vi yeu cau cot loi da co:

- Co mo hinh speaker recognition train/evaluate that.
- Co checkpoint VIVOS phu hop tieng Viet.
- Co SID/SV metrics va artifact.
- Co ung dung assistant tích hop speaker gate.

Fine-tune noisy nen duoc trinh bay la huong mo rong:

He thong hien dung checkpoint ECAPA-TDNN train tren VIVOS. Pipeline fine-tune voi MUSAN/RIR va du lieu nhieu/re da duoc chuan bi trong code, nhung chua dua vao bang metric chinh do gioi han thoi gian/GPU. Day la buoc tiep theo de tang robustness trong moi truong that.

7 Runtime calibration

Ngoai metric training, he thong co benchmark runtime tren audio enroll demo. Ket qua calibration ECAPA:

Backend	Users	Audio	SID Acc	SV Thr	TPR	FPR
ECAPA	5	15	1.000	0.450	1.000	0.000

Calibration nay chi mang tinh demo vi co 5 user, 15 audio, same-session. Same-session co the lam score qua dep so voi thuc te cross-device/cross-session. Threshold 0.45 duoc giu de demo an toan, nhung neu false reject cao trong demo co the can ha threshold va thu lai voi audio that.

8 Web application

8.1 Chuc nang nguoi dung

- Tao user moi va enroll giong noi bang nhieu mau audio.
- Xem va sua profile sau khi unlock bang password.
- Re-enroll voice khi giong thay doi hoac admin yeu cau.
- Them voice samples vao centroid hien tai.
- Xuat du lieu user, xoa user va cascade delete data lien quan.
- Tro ly push-to-talk, text mode va TTS.

8.2 Chuc nang assistant

- NORMAL: hoi gio, hoi thoi tiet demo, ke chuyen cuoi, hoi dap tong quat.
- IMPORTANT: doc ghi chu, gui email, xem so du demo, xoa du lieu, mo file, them ghi chu, them lich, them lien he, dat reminder.
- PERSONAL: chao theo ten, phat nhac theo gu, xem lich ca nhan, liet ke reminder.

8.3 Admin panel

Admin co the:

- Xem thong tin user da masked.
- Duyet yeu cau reset password bang ma/PIN mot lan.
- Yeu cau user re-enroll voice.
- Revoke voice embedding.
- Xem va resolve security alerts.

9 Cac co che bao mat

Co che	Mo ta
Password hashing	PBKDF2-HMAC-SHA256 voi salt ngau nhien.
OAuth security	PKCE, nonce, callback validation, token encryption bang Fernet.
Session cookie	HttpOnly, SameSite=Lax, Secure khi khong chay HTTP dev.
CSRF guard	Request thay doi trang thai can header X-Requested-With.

Co che	Mo ta
Rate limit	Ap dung cho password endpoints, enroll, assistant text, challenge.
Audit log	Ghi JSONL cho cac su kien auth, speaker, admin va assistant.
Strict CSP	Bo inline script khong an toan, dung nonce va self/CDN gioi han.
Path traversal guard	Resolve path va kiem tra nam trong root hop le.
Biometric data	Co tuy chon encrypt WAV at-rest bang Fernet.
Security alerts	Ghi canh bao khi password fail hoac SV fail dang nghi ngo.

10 Testing va reliability

Test suite bao phu cac nhom:

- Path traversal.
- Password hashing va password reset.
- OAuth token encryption.
- Security routes.
- WAV encryption.
- Reminder parsing va notification.
- ASR correction.
- Challenge-response.
- Cascade delete.
- Email flow guard.
- Regression tests cho cac bug da sua.

Lenh chay test:

```
python -m pytest tests/ -q --basetemp .pytest_tmp
```

Neu thoi gian truoc khi nop con han che, nen uu tien demo web va chay cac test focused lien quan bao mat thay vi train lai model.

11 Cac loi da xu ly

Mot so van de thuc te da duoc fix:

- SpeechBrain/HuggingFace symlink issue tren Windows.
- PhoWhisper warning/403 noise khi fetch model.
- NLU nham `general_question` thanh `get_weather`.
- Gemini quota/API key fail gay trai nghiem xau.
- Reminder da set nhung khong hien thong bao.
- Music player dong bi trigger next track.
- VIVOS co WAV loi decode, da filter truoc khi tao split.
- VoxCeleb split mismatch file Kaggle, da tao split tu file ton tai.
- API exception tra HTML lam frontend loi JSON parse, da them JSON error handler.

12 Huong dan chay demo

12.1 Cai dat

```
python -m venv venv
venv\Scripts\activate
pip install -r requirements.txt
Copy-Item .env.example .env
```

Can cai FFmpeg de decode audio tu browser.

12.2 Chay web

HTTP local:

```
$env:FLASK_DEV_HTTP="true"
python -m web.app
```

Mo:

```
http://localhost:5000
```

HTTPS:

```
python web/gen_cert.py
python -m web.app --ssl
```

12.3 Demo flow de quay video

1. Mo `/enroll`, tao it nhat 2 user va record voice samples.
2. Mo `/assistant`.
3. NORMAL: hoi gio hoac ke chuyen cuoi, guest van duoc tra loi.
4. PERSONAL: noi “phat nhac” bang hai user khac nhau, he thong ca nhan hoa.
5. IMPORTANT pass: dung user da enroll noi “doc ghi chu cua toi”.
6. IMPORTANT fail: dung user khac/guest thu doc ghi chu, he thong block.
7. Admin: mo `/admin`, xem masked user info va security alerts neu co.

13 Danh gia viec chua fine-tune noisy

Neu khong kip fine-tune noisy/re, anh huong chinh la robustness trong moi truong co nhieu va re chua duoc chung minh bang so lieu. Tuy nhien, do an van dat phan cot loi:

- Da train/evaluate speaker model tren dataset that.
- Da co checkpoint VIVOS tot cho tieng Viet.
- Da tích hop SID/SV vao tro ly ao co phan quyen.
- Da co bang metric SID, SV, EER, minDCF va threshold.
- Da co caveat ro rang ve domain, split va calibration.

Khi bao ve, nen noi ngan gon:

Phan fine-tune noisy/reverb la huong mo rong da chuan bi pipeline, nhung chua dua vao ket qua chinh vi gioi han thoi gian train. Ket qua chinh hien tai tap trung vao checkpoint VIVOS va speaker-gated assistant.

14 Han che

- Split train/val/test la split noi bo, chua phai official protocol.
- VIVOS la read speech, sach hon moi truong su dung that.
- Runtime calibration co it user va same-session bias.
- Chua co benchmark cross-device, cross-session va noise/reverb day du.
- Gemini NLU/Q&A phu thuoc API key/quota, du da co fallback.
- Gmail/Calendar can OAuth setup dung tren Google Cloud Console.

15 Ket luan

Do an da hoan thien pipeline tu speaker model den secure virtual assistant. Ket qua VIVOS dat SID Top-1 99.31% va SV EER 0.96%, tot hon VoxCeleb Indian trong thiet lap noi bo. Checkpoint VIVOS la lua chon hop ly cho demo tieng Viet.

Ve mat ung dung, he thong da co enrollment, SID/SV gate, assistant voice/text, admin panel, Gmail OAuth, reminder, file manager va nhieu co che bao mat. Phan fine-tune noisy/reverb nen duoc trinh bay la future work, khong nen claim da co ket qua neu chua train xong.

A Cau truc thu muc nen nop

```
Secure-Virtual-Assistant-with-Speaker-Recognition/  
  core/  
  web/  
  cli/  
  training/  
  scripts/  
  tests/  
  checkpoints/vios/  
  report/main.pdf  
  report/main.tex  
  README.md  
  requirements.txt  
  requirements.lock.txt  
  .env.example
```

B Lenh train/evaluate da dung

Train ECAPA:

```
python training/train_ecapa.py \  
  --data_root <speaker-root> \  
  --split_file <iden_split.txt> \  
  --save_dir <save-dir> \  
  --epochs 15 \  
  --batch_size 64 \  
  --lr 1e-3
```

Evaluate SID:

```
python training/evaluate_sid.py \  
  --ckpt <save-dir>/best_model.pt \  
  --spk2idx <save-dir>/spk2idx.json \  
  --data_root <speaker-root> \  
  --split_file <iden_split.txt>
```

Evaluate SV:

```
python training/evaluate_sv.py \  

```

```
--ckpt <save-dir>/best_model.pt \
--data_root <speaker-root> \
--trial_file <veri_test.txt>
```

C Artifact evidence

Folder/File	Noi dung
training/results/vivos/best_checkpoint	Checkpoint VIVOS tot nhat.
training/results/vivos/sid_results.json	SID Top-1 99.31%, Top-5 100.00%.
training/results/vivos/sv_results.json	SV Top-1 99.99%, minDCF 0.0673.
training/results/voxceleb_indian	Evidence baseline VoxCeleb Indian.
checkpoints/vios/best_model.pt	Checkpoint dung cho runtime demo.
checkpoints/vios/spk2idx.json	Mapping speaker class cho checkpoint.